

ARCHITECTURE SELECTION

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Architectural Choice

The kiosk is divided into presentation, application/domain, and infrastructure layers. The Touchscreen UI handles customer interaction; the Order Manager applies ordering rules and coordinates the workflow; Payment Service protects card authorization; and infrastructure adapters isolate menu storage and receipt-printer access. Dependencies flow inward through explicit interfaces, keeping implementation details hidden.

Reason 1 - Clear separation for a small system

The kiosk has one focused ordering flow and a fixed feature set: three coffee types, two sizes, card payment, and receipt printing. A layered design provides strong separation of concerns without the deployment, monitoring, network, and data-consistency overhead of microservices. Each component has one cohesive responsibility and can be tested independently.

Reason 2 - Hardware and data changes remain localized

Printer-driver changes are confined to the Receipt Printer Adapter, while menu or price storage changes remain behind the Menu & Pricing Repository interface. The touchscreen and Order Manager do not need to know device protocols or database details. This reduces coupling and limits the impact of maintenance changes.

Security Advantage

Payment handling is isolated behind the Payment Service interface. Only validated order totals are sent for authorization, card traffic uses TLS to the external processor, and the Order Manager stores only the authorization result rather than raw card data. This narrows the attack surface and supports least privilege at the payment boundary.

Performance Benefit

Local calls between the Touchscreen UI, Order Manager, menu repository, and printer adapter avoid unnecessary distributed-service latency. Menu and pricing data can be cached locally, while only payment authorization requires an external network call. The result is predictable touch response and fast order completion during busy periods.